

CMPT 353 Final Project: Sentiment Analysis on Reddit Comments

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Introduction

Hidden behind a screen, some social media users are comfortable spreading negativity in comment sections, even on positive posts. I personally have noticed an increasing trend in the amount of negativity online: almost every comment section on Instagram, Reddit, TikTok has top comments that are negative. I do not remember experiencing as much of this a few years ago, so I wanted to test whether negativity has increased in comment sections over time.

In this project, I aim to empirically test my perception of negativity online by analysing sentiment trends in Reddit comments. I was inspired by the sentiment analysis on Reddit data listed on the project ideas page. By computing the sentiment score for each comment in a selection of subreddits, I could compare scores over time. This project investigates three core questions:

- Have Reddit comment sections become more negative over time?
- Are some subreddits more negative than others?
- Am I noticing more negativity because negative comments are more popular and thus pushed to the top of comment sections?

Data

I used the Reddit comments corpus available on the cluster. As this is a large dataset, I only extracted 4 months worth of data: April 2019, April 2024, September 2019, and September 2024. March and September were chosen to provide a sample more representative of the year. Sampling from the year itself would have been the most representative of the data, but this would have greatly increased processing time. The data was obtained using the sample code for extraction from the cluster, with slight modifications.

The total count of rows after cleaning and filtering was 6 million. I attempted to sample from this and perform statistical analyses on the cluster but this proved to be very slow. In the end, I decided to sample 10000 rows from each of the three subreddits and locally work on them with python. I chose three subreddits: r/knitting, r/politics, r/movies. Politics is likely to have more negative sentiment, while I'd expect knitting to have more positive sentiment. I chose subreddits that I suspected were different in terms of negativity, so that I could have a bit more of a representative sample of overall sentiment online.

Cleaning

Initial inspection of a small subset of the spark data frame revealed many comments not helpful in my analysis: I filtered out comments with the author AutoModerator, as these are comments from moderators who remove comments that violate subreddit rules. Next, I noticed that deleted comments or accounts would have a body "[deleted]" or "[removed]", so those were filtered out. Some of the longer comments were just links, which are not helpful in sentiment analysis. Many of the values in the up votes column were missing, so I removed those when determining correlation between sentiment and upvotes. Regarding outliers, since negative Vader scores are relatively evenly spread between -1 and 0, we do not have any numerical outliers to remove.

Sentiment Analysis

As I researched sentiment analysis online, I found Vader: a pre-trained model that "uses a lexicon of sentiment-related words to determine the overall sentiment of a given body of text" (Malde, 2020). The scores range from -1 to 1, with a more negative score correlating with more negative sentiment, and a more positive score with positive sentiment. Vader is designed for social media text: where capitals and punctuation matter. In other sentiment analyses, words are often converted to lower case and punctuation and emojis are removed. But in the case of social media comments "!!!" or "YES", and emojis can convey sentiment and Vader takes this into account (Malde, 2020). Vader also deals with stop words, so I

did not remove those. I applied Vader's sentiment intensity analyser to the data column with the body of each comment and created a new column called vader_score to store the sentiment scores.

Initial Visualization

Figure A is a plot of the distribution of sentiment scores by subreddit. It seems there are many more neutral comments in general. Aside from r/knitting which seems to be a very positive subreddit, the amount of negativity and positivity in comment sections is about equal, and most comments are neutral. This could support the idea that the perception of increased negativity online could be due to negative comments being filtered to the top of comment sections to increase engagement; comment sections are not generally negative, but negativity is popular.

Figure A:

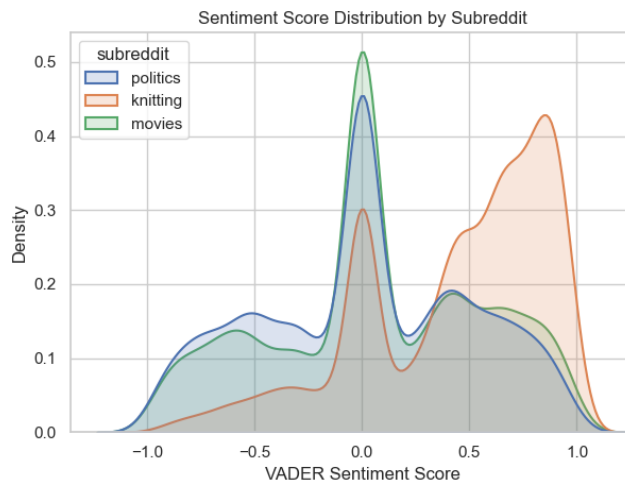
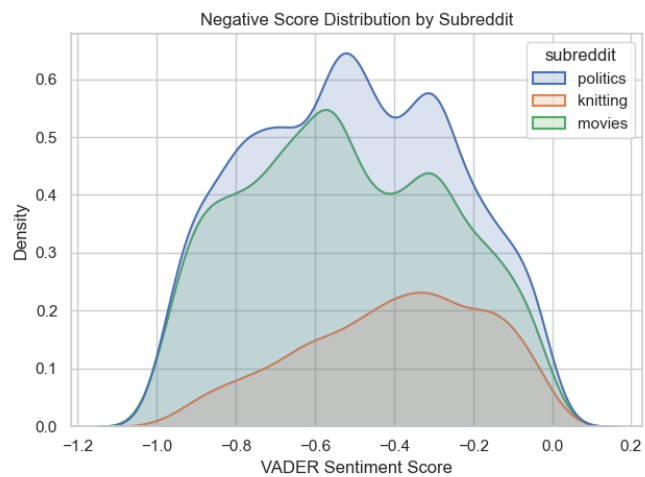


Figure B:



Since the focus of this project is focused negativity in comment sections, I created a subset of just the negative scores, as seen in Figure B. This subset will be used in the rest of the project.

Statistical Analysis

Data Assumptions

Initially, I wanted to compare average sentiment between 2019 and 2024 to see if subreddits in general have gotten more negative. The appropriate statistical test for this query is a t-test, which determines whether the two groups' scores have similar means. This test requires the data to be normally distributed with the two groups also being identically distributed (having similar variance). I used a normality test and an equal variance test (Levene tests), both of which have null hypotheses that the data is normally distributed and has equal variance among the groups. The tests returned p values less than 0.05, so the negative Vader sentiment scores failed to meet these assumptions. Thus, analysis of the data had to be performed with nonparametric tests that do not have assumptions about the distribution of the data.

Statistical Techniques

This project employs four nonparametric tests. Three of these tests return p values which give the probability of observing our data given that our null hypothesis is true. If p is very small, it is likely that our null hypothesis is not true, so we can accept the alternative hypothesis.

The first test is a Mann-Whitney U test which tests whether the observations in one group tend to be ranked higher than those in the other group when all of the data is sorted. The test returns a p value, and if it is less than 0.05, we can conclude that one group's values are higher than the other group's. To compare the three subreddits, I used the Kruskal Wallis test, which is an extension of the Mann-Whitney U test that

can be applied to multiple groups at once (*Kruskal-Wallis H Test*, n.d.). The test assumes that the groups' distributions have the same shape. Figure B above is a plot of the distribution of negative Vader scores for each group, showing that the distribution shapes are not too dissimilar, so the results of this test on the data can be trusted. The Kruskal Wallis test tells us whether or not at least one of the groups has values that tend to rank higher than other groups' values. Once again, it returns a p value, which if very small, indicates that one group's values rank higher than the others. Dunn's test is then used with a Bonferroni correction to determine which groups are different (*A post-hoc test for Kruskal-Wallis*, 2023). This returns a p value for each pair of groups indicating whether one group's values rank differently.

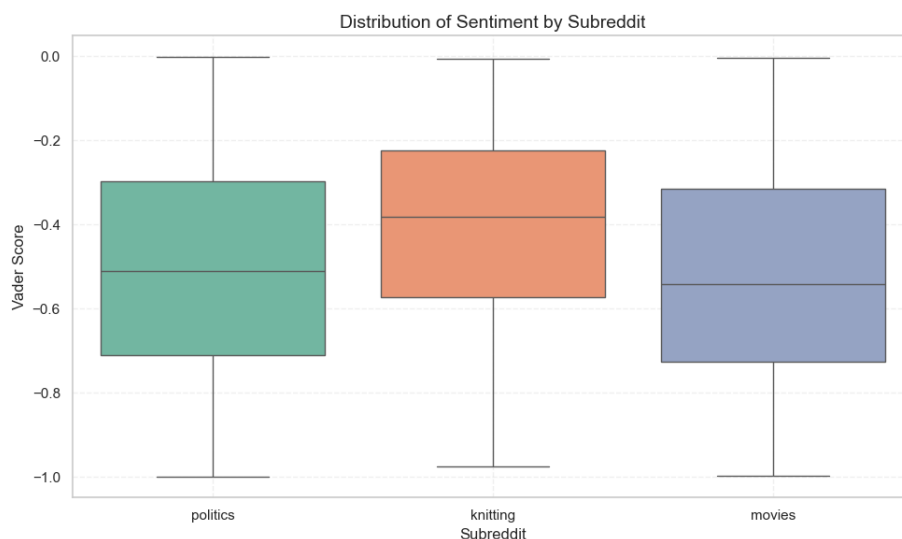
Lastly, I use the Spearman's Rank Order Correlation to determine if there is a relationship between the number of upvotes a comment receives and its degree of negativity (*Spearman's Rank-Order Correlation*, n.d.). This tests whether two variables increase or decrease together. It returns a value between -1 and 1, indicating the strength and direction of the relationship: -1 indicates that the variables are strongly negatively related, so if one variable increases, the other decreases. A score of 1 indicates the variables are strongly positively related. This is not necessarily the same as a linear relationship, as the test can capture trends that do not follow a straight line when plotted.

Results

The Mann-Whiney U test for the negative Vader scores from 2019 and 2024 did not produce a significant result ($p > 0.05$). So, we can not conclude that one year's scores are higher or lower than the other year's scores: the ranking of scores is not significantly different between the groups.

Since a sentiment comparison overtime shows no obvious differences, I moved on to testing which subreddit was more negative. I performed a Kruskal Wallis test on the negative Vader scores for the three subreddits over the two years. This resulted in a very small p value ($p < 0.05$), suggesting that at least one of the groups' Vader scores are often ranked higher or lower than the others. This means that at least one subreddit tends to have higher or lower sentiment scores than the other subreddits, which is equivalent to saying that one subreddit is more or less negative than the others.

Figure C:

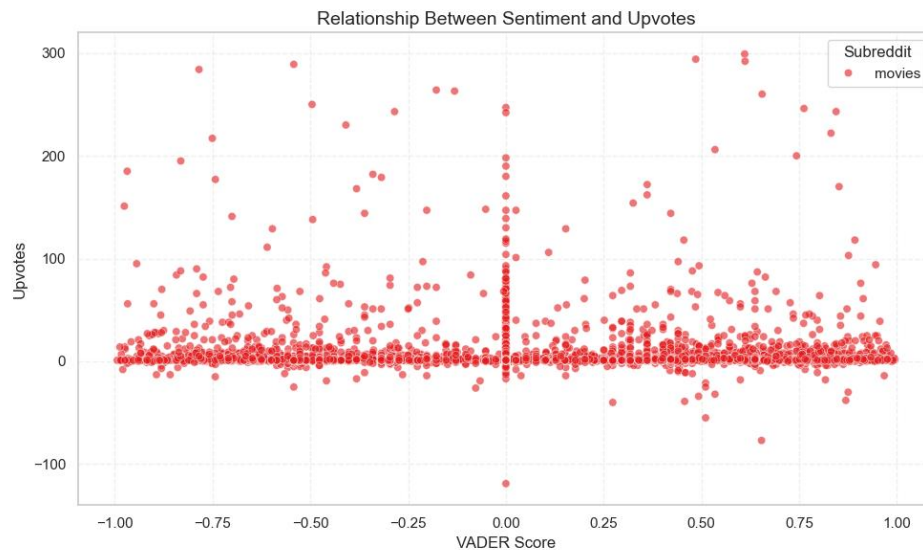


To determine exactly which groups were different, I performed Dunn's test on the negative scores for all three subreddits and observed p values less than 0.05 for all pairs of subreddits, revealing that all three subreddits have different sentiment rankings. To determine which subreddit was most negative, I attempted to create box plots to see if the spread of a subreddit's scores skewed more negative. However,

because sentiment scores are very small (between -1 and 0) and we have a lot of data, any differences are difficult to see on the graph above. Thus, I calculated the median sentiment scores as follows: knitting = -0.3818, movies = -0.5423, politics = -0.5106. I used the median rather than the mean because the tests used on the data compare the ranking of values not their magnitude. As expected, based on Figure A on page 2, knitting is much more positive than the other two subreddits. Politics and movies are similar in terms of negativity but movies has slightly more negative comment sections. These results could suggest that subreddits where people express opinions are likely to be more negative, which is intuitive.

Now that we know r/movies is more negative, I would like to see if the number of upvotes are related to the negativity score of a comment in this subreddit. The Spearman correlation was calculated as 0.0029, so there is no significant relationship between upvotes and negativity. A scatter plot of this below confirms there is no discernable trend in the data.

Figure D:



Conclusion

This project set out to test whether Reddit comments have become more negative over time, whether certain subreddits have more negative comment sections, and whether negativity is linked to comment popularity. Using Vader's sentiment scores on comments from 2019 and 2024, I determined that overall negativity in Reddit comment sections has stayed relatively stable over time, answering my first question. Some subreddits were notably more negative than others, and although I did not test for the following, I assume this is due to some subreddits being more opinion driven. Finally, negativity did not appear to influence the popularity of a comment, so the hypothesis, that my perception of negativity differs from the true negativity of comment sections because these comments are more popular, is likely false for Reddit. The opposite could be true for other social media platforms with different algorithms for comment ranking.

Limitations

Dealing with big data proved to be time consuming and difficult. I experienced a few setbacks and found things were taking longer than expected. A major challenge involved spark incompatibility with statistical tests. I wanted to perform variance and normality tests on the data from the two different years but could not find Spark functions to do this. Collecting the entire dataset into python to use stats.Levene or stats.normaltest was infeasible due to the size of the data. Even sampling a small fraction of the data to test with was very slow.

I initially wanted to compare sentiment between Vancouver and Toronto, as I have heard Toronto is a friendlier city than Vancouver. I realised after extraction and initial analysis that the subreddits r/Vancouver and r/Toronto were not retrieved from the cluster data set. However, due to time constraints, I could not redo my extraction. In hindsight, I should have grouped by subreddit to inspect the data in spark before extracting it.

With more time, I would have liked to sample from some larger subreddits like AskReddit or TodayILearned as I feel that these would have more people giving their opinions in the comments. However, these subreddits are so popular that extracting the data would not have been feasible in terms of time. Additionally, an interesting analysis would have been to find subreddits that were more negative and determine what the most common topics were in that community: perhaps we could figure out what topics attracted more negativity online. Lastly, testing differences over a 5 year period could be why the negativity seems to have remained stable over time. Exploring sentiment differences over a longer period of time could reveal significant differences in sentiment.

References

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- Spearman's Rank-Order Correlation - A guide to when to use it, what it does and what the assumptions are*. (n.d.). Laerd Statistics. <https://statistics.laerd.com/statistical-guides/spearmans-rank-order-correlation-statistical-guide.php>

Accomplishment Statements

- Implemented an Extract-Transform-Load pipeline using PySpark on a compute cluster to preprocess over 6 million Reddit comments, transforming and reducing dataset size for efficient analysis in Python
- Leveraged NLTK's Vader to identify sentiment trends, subreddit-level patterns, and determine if a relationship exists between negativity and comment popularity
- Analysed raw data to determine and conduct the appropriate statistical analyses (Mann-Whitney U, Kruskal-Wallis, Dunn's test) based on data distributions and project objectives to compare sentiment between years and subreddits